

Robert Chen

robert@robertwchen.com | (240) 898-8894 | robertwchen.com | github.com/robertwchen | linkedin.com/in/robertwchen

EDUCATION

University of Virginia | GPA: 3.6/4.0

Charlottesville, VA

B.S. in Computer Engineering & Electrical Engineering

May 2026

- Relevant Courses: Embedded Computer Systems, Autonomous Robots, Signals and Systems, Machine Learning

PROJECTS

Badminton Swing Consistency & Deception Analyzer | *Flutter, Dart, Embedded C, BLE, IMU, PCB Design, DSP, SQLite*

- Built an end-to-end racket-mounted sensing system for real-time swing speed and impact feedback by integrating Embedded C firmware, BLE telemetry, a custom 2-layer PCB, Flutter app, SQLite storage, and enclosure design
- Achieved ~**95%** hit-vs-shadow-swing classification accuracy in on-court tests by designing **FFT-based** and physics-based features from 100 Hz IMU and audio data
- Designed and routed a 2-layer PCB and verified I2C sensor bring-up with logic analyzer traces

Vision-Assisted Autonomous Quadcopter | *ArduPilot, MAVLink, Crossfire, UART, OpenCV, Power Regulation*

- Converted an FPV racing quad into an **autonomy-capable** platform by deploying ArduPilot on an H7 flight controller, configuring CRSF UART manual override, and validating flight logging
- Built a companion vision-control loop that detects an AprilTag and streams MAVLink velocity setpoints in Guided mode; added watchdog and a hardened **5V power path** to ensure safe takeover on target loss

Programmable 32-bit RISC Processor | *Verilog, RTL Design, Simulation, Verification*

- Implemented a 32-bit RISC CPU in **Verilog** (ALU, register file, control, decode) and iterated the control path to match expected instruction behavior
- Created test vectors and debugged via **waveform simulation** to verify execution across normal and edge cases

EXPERIENCE

Research Assistant (ML & Computer Vision)

Charlottesville, VA

Visual Intelligence Lab, University of Virginia

Sep 2025 - Present

- Developed a Python computer vision (CV) and OCR pipeline (**OpenCV, Tesseract**) for tennis broadcast score overlays, outputting JSON match state with temporal smoothing and a finite-state machine
- Building a reproducible Python data pipeline (pandas, NumPy) and defining a Serve Performance Index using speed, outcomes, and 3D keypoints for consistent evaluation criteria

DevOps Cloud Automation Engineer Intern

Fairfax, VA

Verint

May 2025 - Nov 2025

- Migrated Terraform workspaces to **OpenTofu**; standardized remote state and reusable modules for EC2, S3, and IAM, reducing plan/build times and removing vendor lock-in
- Delivered an **AWS Transfer Family** web client on ECS Fargate (DynamoDB, AWS S3, Amazon CloudFront), with Docker, CloudWatch, and an Angular front end for secure, compliant file transfers
- Engineered CI/CD with **GitHub Actions** and Harness, automating PR comments and checks (Python, Bash), establishing ~**2-3 min** time-to-feedback on ~**30 PRs per week** and removing manual checks

Software Engineering Intern

Centreville, VA

S&J Chen, LLC

May 2024 - Aug 2024

- Shipped a full-stack tenant payment portal for 2 shopping centers (Next.js, TypeScript, Supabase, Vercel) with admin tools for tenants, leases, and invoices, cutting manual rent follow-ups by ~**60%**
- Integrated **Stripe webhooks** with reconciliation to auto-update invoices and drive CSV reporting, saving ~**4 hrs/month** and reducing monthly reporting from ~**3 hours** to ~**30 minutes**

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Verilog, MATLAB, Bash, SQL, Java, JavaScript/TypeScript, HTML/CSS

Embedded & Robotics: Embedded Linux, firmware, device drivers, testing/validation, UART, I2C, SPI, BLE, ArduPilot, Git

Hardware and Tools: PCB Design, schematic capture, KiCad, Altium, Cadence Virtuoso, Autodesk Inventor

ML & Perception: OpenCV, PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

LEADERSHIP & ACTIVITIES

Leadership: Yo-Yo Club, President (founder, ~50 members); Taiwanese Student Association (TSA), Sports Chair

Interests: Tennis, Volleyball, Pickleball, Figure Skating, Robotics